Variant Calling and Benchmarking of HG002 Whole-Exome Sequencing Data

Group 5

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Workflow

Data preparation and alignment

The analysis began with the GIAB HG002 WES BAM file, which had been aligned to an earlier reference build. To ensure consistency with GRCh38, the BAM was converted to paired-end FASTQs and realigned. Because the original BAM was coordinate-sorted, it was first name-sorted prior to BAM-to-FASTQ conversion to preserve paired-end reads. The resulting FASTQs were then realigned to the GRCh38 reference genome using BWA-MEM. Duplicate reads were marked using Picard MarkDuplicates, and the deduplicated BAM was subset to chromosome 22.

Coverage estimation and downsampling

Coverage estimation was performed using the Agilent SureSelect V5 exome capture BED file downloaded from Agilent's website to obtain representative depth estimates for WES data. Exome-restricted depth on chr22 yielded a baseline coverage of approximately 215× across ~1.09 million targeted bases, enabling informed downsampling to multiple validated coverage levels.

Table. Mean coverage depth on chromosome 22 calculated using genome-wide versus exome-restricted methods

Dataset	Depth calculation method	Mean depth on chr22
Baseline (no BED)	Genome-wide chr22	~14.3×
Baseline (exome-restricted)	Agilent SureSelect V5 BED	~215×
Downsampled 80x	Exome-restricted (target ~80×)	~79.8×
Downsampled 40x	Exome-restricted (target ~40×)	~39.9×
Downsampled 10x	Exome-restricted (target ~10×)	~10.0×
Downsampled 2x	Exome-restricted (target ~2×)	~2.0×

Base quality score recalibration and variant calling

Known variant sites from dbSNP138 were used for base quality score recalibration with 'GATK BaseRecalibrator' and 'ApplyBQSR', with 'AnalyzeCovariates' used for quality assessment. Variants were called using 'GATK HaplotypeCaller' in GVCF mode restricted to chr22, followed by joint genotyping with 'GenotypeGVCFs' to produce per-coverage VCF files annotated with dbSNP rsIDs.

Variant stratification and filtration

Each VCF was split into SNP and INDEL subsets using GATK SelectVariants. Quality metrics including QD, MQ, MQRankSum, FS, SOR, ReadPosRankSum, DP, and InbreedingCoeff were examined using density plots stratified by coverage, variant type (SNPs and INDELs), and variant novelty (rsID-annotated versus novel). These plots informed coverage-aware hard-filter thresholds applied separately for SNPs and INDELs using GATK VariantFiltration. Maximum DP thresholds were computed per coverage level as mean(DP) + 5×SD(DP) within exome capture regions to account for coverage-dependent scaling. Summary reports captured total, PASS, and filtered variants, as well as per-filter-tag contributions stratified by known and novel variants.

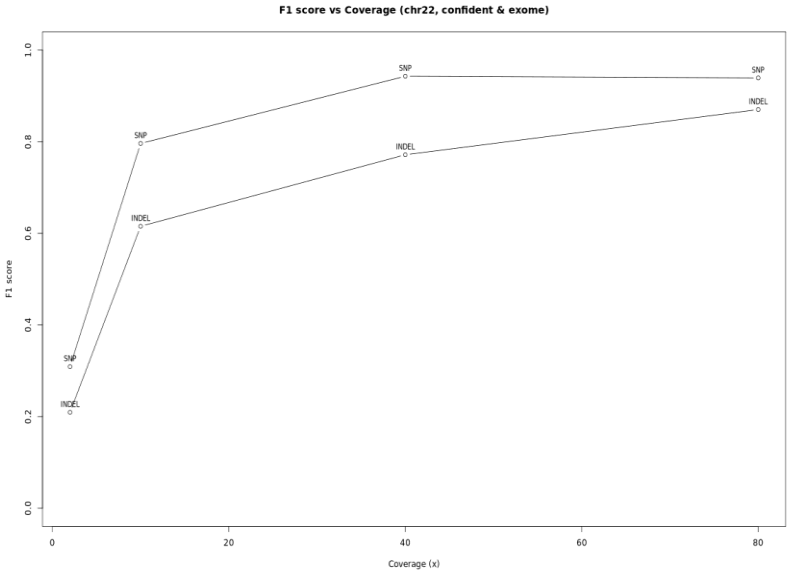
Benchmarking against GIAB truth data

Benchmarking was performed using the GIAB HG002 GRCh38 truth set restricted to chr22 and intersected with the exome capture BED to limit evaluation to confident, assay-relevant regions, as recommended in '*Best practices for benchmarking germline small-variant calls in human genomes'*. Benchmarking was conducted using hap.py with the 'vcfeval' engine in a dedicated Conda environment, with metrics extracted including precision, recall, F1 score, genotype concordance, and false positives per megabase.

Variant Calling Results

Quality metric exploration showed clear separation between known and novel variants across coverage levels. At low coverage, novel variants were enriched in low-quality metric tails, while higher coverage produced tighter and more favorable distributions. Coverage-aware hard filtering preferentially removed novel variants while largely preserving known variants.

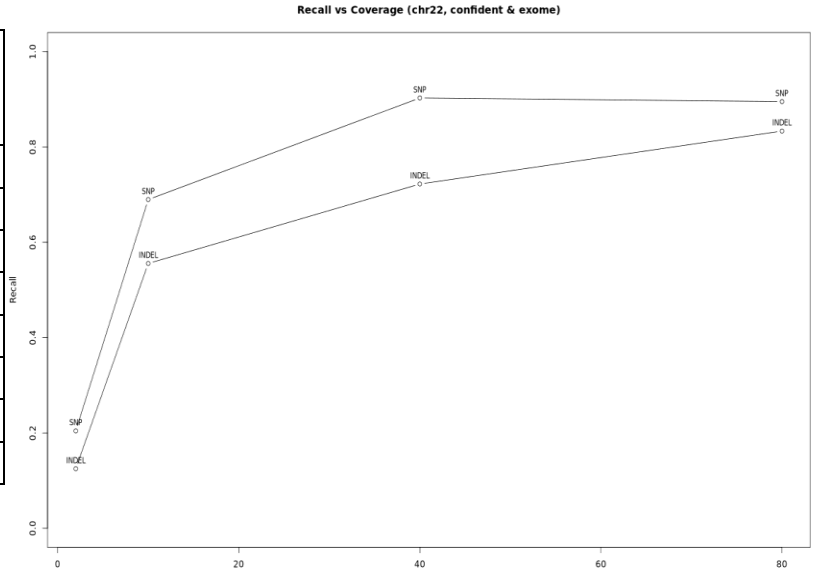
The number of rsID-annotated variants increased with coverage, indicating improved sensitivity of variant detection and motivating the downstream benchmarking analysis.

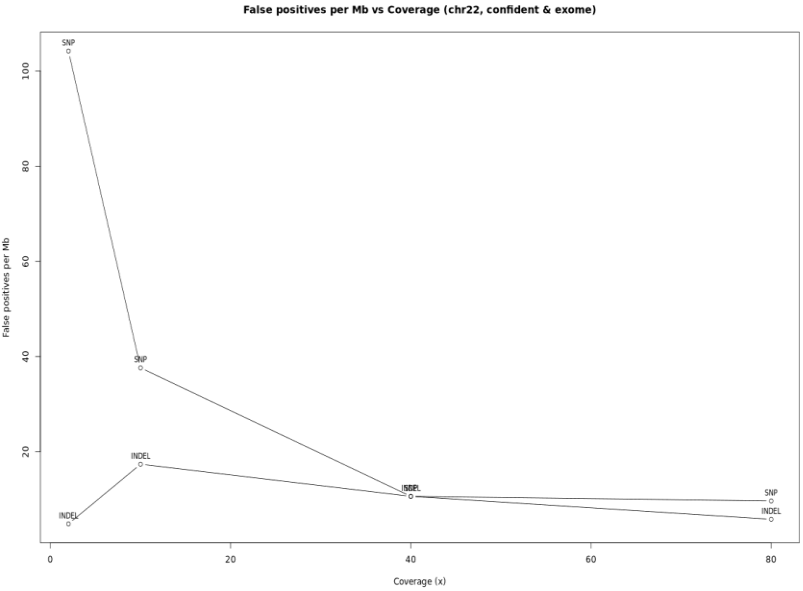


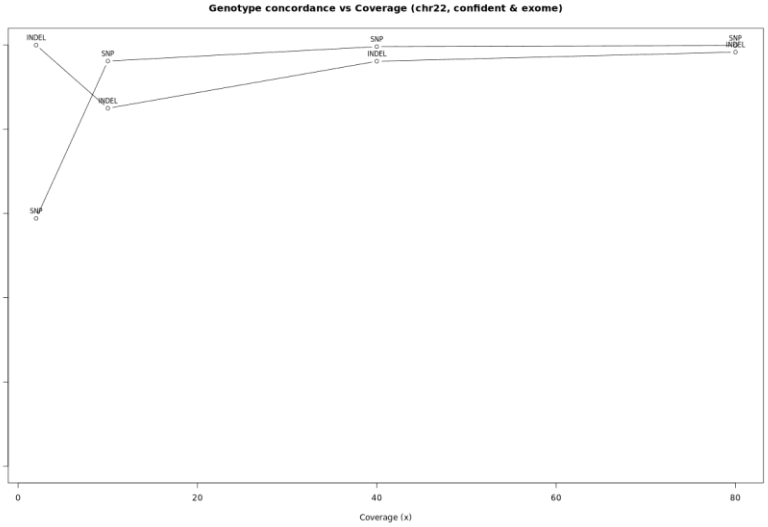
Benchmarking Results

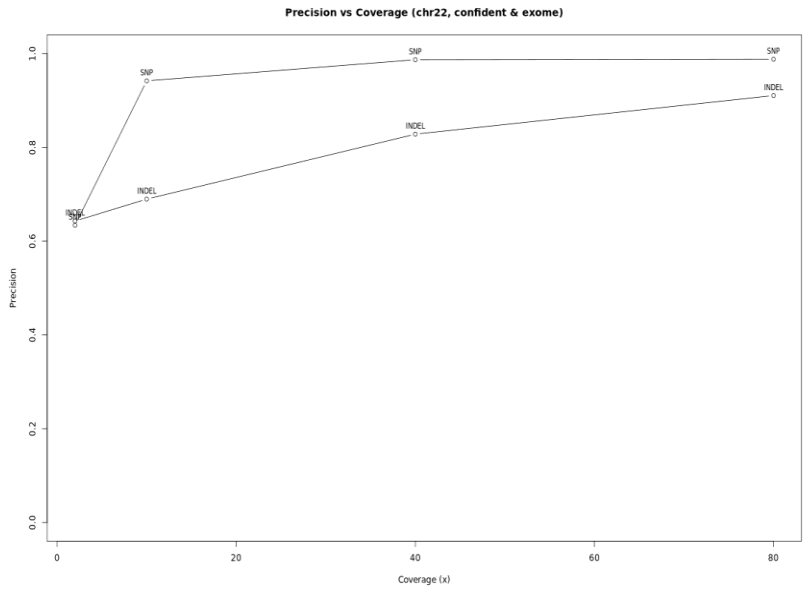
Benchmarking results are summarized across four coverage levels (2×, 10×, 40×, and 80×) and stratified by variant type.

Coverage	Variant type	Precision	Recall	F1 score	Genotype concord	False positives per Mb
2×	SNP	0.6	0.	0.3	0.5882	104.21
	IND	0.6	0.	0.2	1.0000	4.82
10×	SNP	0.9	0.	0.7	0.9620	37.63
	IND	0.6	0.	0.6	0.8500	17.37
40×	SNP	0.9	0.	0.9	0.9964	10.61
	IND	0.8	0.	0.7	0.9615	10.61
80×	SNP	0.9	0.	0.9	0.9988	9.65
	IND	0.9	0.	0.8	0.9833	5.79









Interpretation of Trends

Effect of depth of coverage on accuracy

Restricting depth calculations to exome capture regions is essential for interpreting WES benchmarking results. When depth is averaged genome-wide across chr22 (approximately 131 Mb), the apparent coverage ~14× substantially underrepresents the effective depth within targeted regions. Restricting depth calculations to the ~1.09 Mb of exome capture targets reveals the true sequencing depth ~215× and explains the strong performance gains observed when moving from 2× to 10× coverage, as well as the diminishing returns beyond ~40×. Interpreting coverage in this assay-relevant context is essential for understanding the observed trends in recall, genotype concordance, and False Positives per Mb.

Increasing coverage depth substantially improves variant detection accuracy, particularly recall and genotype concordance. The most pronounced improvement occurs when increasing coverage from very low depth (2×) to moderate depth (10×), where recall rises sharply as more true variants receive sufficient read support to be detected and confidently genotyped. Beyond this point, additional increases in coverage continue to improve accuracy but with diminishing returns. At higher coverage levels (40× and 80×), recall and genotype concordance approach saturation, especially for SNPs, indicating that most callable variants within the exome capture regions are already being reliably detected and correctly genotyped.

Trade-off between sequencing cost and accuracy

While higher coverage improves accuracy, the gains diminish beyond approximately 40×. This suggests that extremely high coverage may not be cost-effective for WES studies when balanced against sequencing costs.

Does higher depth always improve performance?

Higher depth does improve performance, but not linearly. Precision, recall, and F1 score show diminishing returns at higher coverage levels, indicating a practical upper bound beyond which additional sequencing yields limited benefit.

Supporting Files

Supporting files include the unified benchmarking summary TSV file (benchmarking_summary_chr22.tsv), the bash workflow script used in the analysis (provided as Project_Script_Group5.sh), supplementary density plots for variant quality metrics (provided as Density_plots.zip), SNP and INDEL filter summary reports (provided as SNP_INDEL_filter_reports.zip), all benchmarking plots generated by hap.py (precision, recall, F1 score, genotype concordance, and false positives per megabase), provided as benchmarking_plots.zip, a visual infographic summarizing the complete variant calling and benchmarking workflow (provided as HG002 WES Variant Calling and Benchmarking Workflow.pdf), and a separate technical log documenting troubleshooting steps and iterative corrections throughout the project (provided as Technical Journey - Variant Calling and Benchmarking Pipeline Development.pdf).